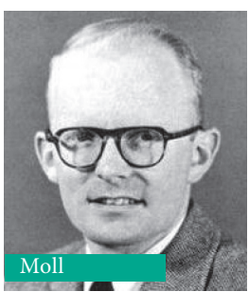
Table 6.3 Comparison of JFET and MOSFET

S.No	JFET	MOSFET
1.	JFET stands for Junction Field Effect Transistor	MOSFET stands for Metal Oxide Semiconductor Field Effect Transistor
2.	It can be operated only in the depletion mode	It can be operated either depletion or in enhancement mode
3.	It has high input impedance (10^8 Ω)	It has very high input impedance (from 10^{10} Ω to 10^{15} Ω)
4.	It is difficult to fabricate	It is easier to fabricate
5.	Gate current (Ig) is high	Gate current (Ig) is low
6.	It is mainly used in low noise application	It is widely used in VLSI circuits

6.11. SILICON CONTROLLED RECTIFIER (SCR)

SCR is a three terminal and three junction semiconductor device acts as true electronic switch. It is a unidirectional device. It converts AC to DC and controls the amount of power fed to the load. It contains the features of a rectifier and transistor. SCR is widely used device in the Thyristor family, so it is commonly called as Thyristor.


Moll

History: The SCR was proposed by William Shockley in 1950 and championed by Moll and others at Bell Lab. It was developed by Gordon Hall and his team engineers at General Electric Company in the year 1956.

Construction: SCR consists of four semiconductor layers forming a PNPN-structure as shown in the Figure 6.15. There are three junctions namely J_1, J_2, J_3. SCR have three leads, they are anode (A), cathode (K) and gate (G). The end P-layer acts as anode, the end N-layer acts as cathode and the P-layer nearer to cathode acts as gate.

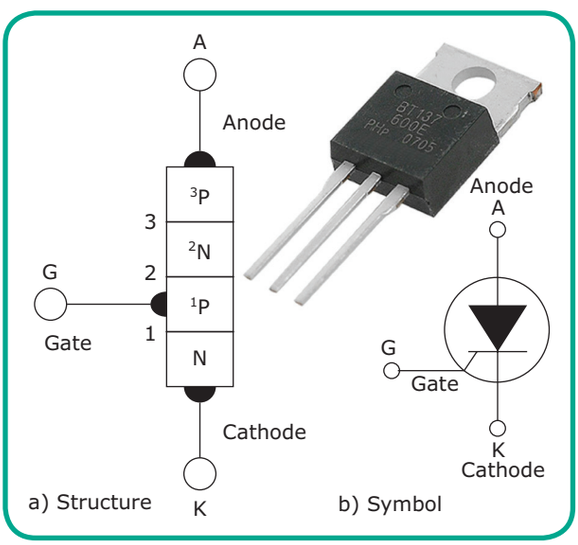


Figure 6.15: **SCR Structure and Symbol**

Working Principle: In the normal operating conditions of SCR, the anode (A) is always kept at high positive potential with respect to cathode (K), and gate (G) is at small positive potential with respect to cathode. A load resistor (R_L) is

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